Alcohols

Ingyu Bahng

DATE PLACEHOLDER

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1 Alcohol Nomenclature & Foundations

Alcohols are a fundamental class of compounds characterized by the presence of one or more hydroxyl (– OH) groups attached to a carbon atom.

Naming alcohol is done through the following steps.

(1) Find the longest chain **containing** the – OH group and switch the suffix of alkan**e** to alkan**ol**. When there are multiple – OH groups, the the suffix -ol becomes -diol, triol, etc. *Mind that the chosen chain may not be the longest chain in the molecule.*

(2) Number from the end **closest** to the – OH group.

(3) Follow the rest of the naming conventions in alkanes.

Here are some examples of alcohol names using the nomenclature steps above.

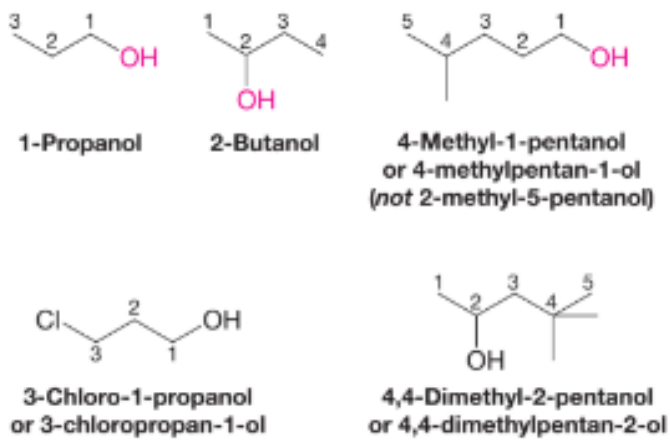

Figure 1: Alcohol Nomenclature Examples

Cyclic alcohols are called **cycloalkanols** and follow the same nomenclature system as cycloalkanes.

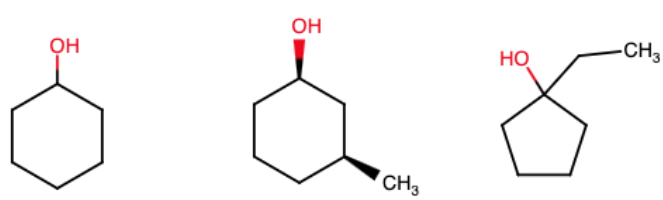

Figure 2: Cycloalkanol Nomenclature Examples

The names of the cycloalkanols above are

cyclohexan-1-ol cis-3-methyl-cyclohexan-1-ol 1-ethyl-cyclopentan-1-ol

Also, in cases where $-OH$ or $-OR$ groups are defined as substituents, they are called hydroxy- and alkoxy-, respectively. *This is analogous to alkanes being called alkyl-.*

Just like we have learned about primary, secondary, and tertiary halogen reactions, we also use the same system to define alcohol molecules as well.

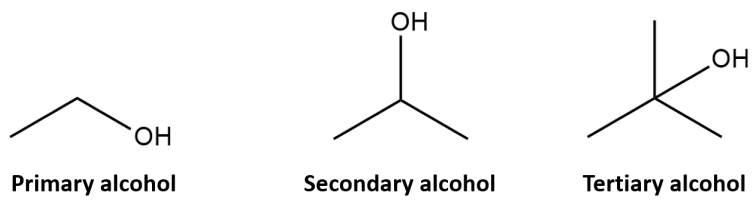

Figure 3: Primary, Secondary, and Tertiary Alcohols

Let's talk about the physical properties of alcohols. Alcohols have relatively **high melting and boiling points**, and **water solubility** because of their ability to hydrogen bond, increasing IMFs and allowing solvation by water.

Keep in mind that water solubility decreases with the size of R because **R is hydrophobic** and **-OH is hydrophilic**.

Let us take a look at the acidity across various alcohols.

$$ROH + H_2O \rightleftharpoons RO^- + H_3O^+$$

Using the shell reaction above, here are the pK_as of alcohols across various R groups. As a reminder, lower pK_a values means higher acidity.

#	ROH	pK_a
1	H_2O	15.7
2	CH_3OH	15.5
3	CH_3CH_2OH	15.9
4	$(CH_3)_2CHOH$	17.1
5	$(CH_3)_3COH$	18
6	$ClCH_2CH_2OH$	14.3
7	CF_3CH_2OH	12.4
8	$CF_3CH_2CH_2OH$	14.6

Recognize that as the alcohol shifts from methyl → primary → secondary → tertiary R groups (#2 → #3 → #4 → #5), the acidity of the alcohol decreases. This pattern is primarily driven by **solvation**. In water or alcohol solvents, the negative charge on an alkoxide ion (RO^-) is stabilized through solvent molecules surrounding the $-O^-$ to spread out the charge.

Higher acidity when R contains more halogens (#7) compared to less (#6) occurs due to the **inductive effect**. The resulting RO^- is more stablized when the R group is able to share the burden of the concentrated negative charge on the O. With more halogens (*which are very EN elements*), the negative charge is more easily spread out. This effect tapers off with increasing distance between the halogens and the $-O^-$ as shown in #8.

As a review of past concepts, remember that these pK_a values are all relative, meaning that under varying conditions, the same alcohol can act as a base or acid (**amphoteric**). For instance, when methanol is mixed with HCl, the resulting mixture will treat methanol as a base because HCl has a much lower pK_a value than methanol.

$$CH_3OH + HCl \rightleftharpoons CH_3OH_2^+ + Cl^-$$

In the case where we mix methanol with a higher relative pK_a value (*such as a strong base*), methanol would be treated as an acid.

$$CH_3OH + NaOH \rightleftharpoons CH_3O^- + Na^+ + H_2O$$

Recall that the mixture above is a pH buffer solution—a mixture of an weak acid or base with its cooresponding conjugate base or acid. We have covered this topic in *Introductory Chemistry 2.3.3 Titration and Buffers*.

2 Alcohol Synthesis

Alcohol is synthesized primarily through one of three different methods. We will go through each method in turn.

Firstly, through **RADICAL HALOGENATION AND SN2/1**. Our goal here is the following

$$R-L + OH^- \longrightarrow R-OH + L^-$$

When we have H_3C-L or $R_{prim}-L$ only, we can accomplish this reaction purely through SN2 because there is little hinderance around the electrophile.

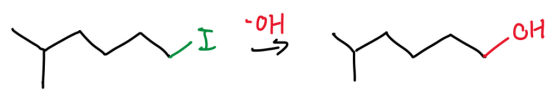

Figure 4: Alcohol Synthesis via SN2

However, the possiblity of E2 reactions is introduced when β branching is present

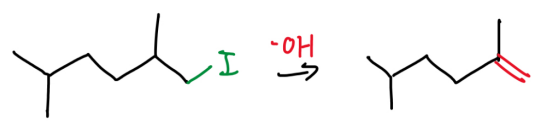

Figure 5: E2 Competition in Primary Alcohols with Beta Branching

We can resolve the E2 problem by using a less basic, "masked " OH-equivalent such as acetate.

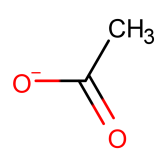

Figure 6: Acetate Ion Structure

After substituting the halogen leaving group in a β branched R group, we can convert the acetate subtitutent into a -OH group via hydrolysis.

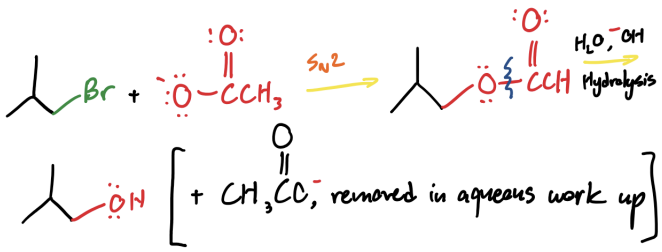

Figure 7: Alcohol Synthesis via Acetate Substitution and Hydrolysis

Moving onto $R_{sec} - L$ and $R_{ter} - L$, SN2 is inefficient or impossible due to steric hinderance around the electrophile, thus we carry out alcohol synthesis via SN1. By using a poor Nu like H_2O, we can avoid E2. However, again, there is the possiblity of E1 reactions. The only solution to this is to carry out the reaction at lower temperatures to minimize the free energy push the E1 reaction gets from having a greater entropy change (ΔS) than the SN1 reaction.

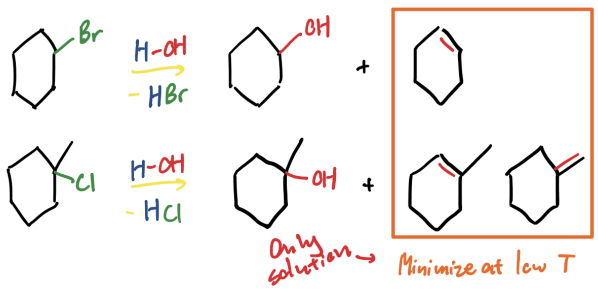

Figure 8: Alcohol Synthesis via SN1

Secondly, alcohol can be synthesized via the **REDUCTION OF ALDEHYDE AND KETONES**.

Figure 9: Aldehyde and Ketone Structures

At the end of the day, what we are trying to do is incorporate H_2 into the molecule to produce a $R_{sec} - OH$ or $R_{prim} - OH$ alcohol.

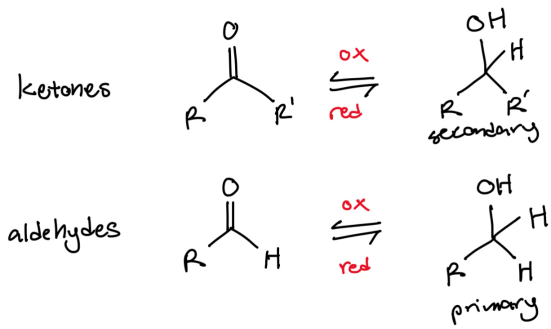

Figure 10: Reduction of Aldehydes and Ketones to Alcohols

Direct hydrogenation is possible, but cumbersome because it needs H_2 pressure and catalysts (e.g. Rd, Pt). Instead, the most common laboratory method is via **hydrides** (H^-). Hydride donors are prepared either through boron (B) or aluminum (Al) via the following reactions.

$$Na^+H^- + BH_3 \longrightarrow Na^+BH_4{}^- \qquad \textit{sodium borohydride}$$

$$Li^+H^- + AlH_3 \longrightarrow Li^+AlH_4{}^- \qquad \textit{lithium aluminum hydride}$$

These reagents ($BH_4{}^-$ and AlH_{40}) are still basic by nature and will undergo protonation by alcohols or water (protic) to give H_2, which is not what we want.

However, it's important to keep in mind that the reactivities of $AlH_4{}^-$ and $BH_4{}^-$ differ. $AlH_4{}^-$ is a much more aggressive hydride donor compared to $BH_4{}^-$ because Al is a less electronegative element compared to B, thus the added negative charge near from the hydride is more unstable compared to B. Consequently, $AlH_4{}^-$ is so reactive that it will react even with air moisture to form H_2.

Thus the $Na^+BH_4{}^-$ reaction with water is relatively slow, **allowing us to use alcohol solvents** if needed without worrying about protonation.

$$Na^+BH_4{}^- \xrightarrow{H_2O} 4\,H_2 + Na^+B(OH)_4{}^- \qquad \textit{slow}$$

On the other hand, $Li^+AlH_4{}^-$ is much more aggressive as we have explained above, thus we must use aprotic or dry solvents to prevent protonation, such as ethers.

$$Li^+AlH_4{}^- \xrightarrow{H_2O} 4\,H_2 + Li^+Al(OH)_4{}^- \qquad \textit{violent}$$

Definition 2.1: placeholder

$Li^+AlH_4{}^-$ (but not $Na^+BH_4{}^-$) will reduce even halides.

$$RX + LiAlH_4 \longrightarrow R-H$$

The donated hydride will thus attack the central carbon of the ketone or aldehyde, resulting in an alkoxide.

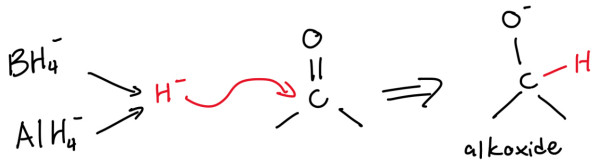

Figure 11: Formation of an Alkoxide Intermediate

The resulting alkoxide is then protonated by alcohol solvent (in the case of $Na^{+}BH_4^{-}$) or aqueous work up (in the case of $Li^{+}AlH_4^{-}$). Mind that the aqeuous work up is in a separate step after purification.

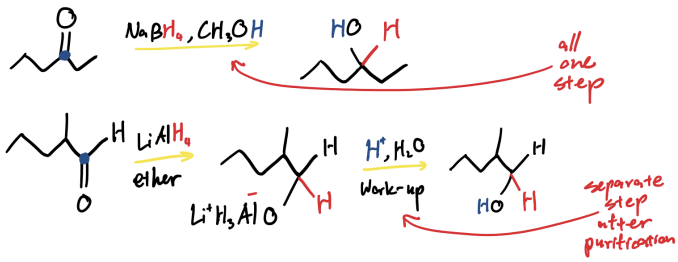

Figure 12: Protonation of Alkoxide to Form Alcohol

In all, the mechanism of $NaBH_4$ is termolecular, concerted, and simultaneous. On the other hand, $LiAlH_4$ is a stepwise delivery of [1] H^- and [2] H^+, H_2O (the work-up step).

We can **oxidize these resulting alcohols back to aldehydes and ketones** by using Cr^{VI} reagents, which accept e^-s and consequently form Cr^{III}. One such compound is chromic acid, formed via the following.

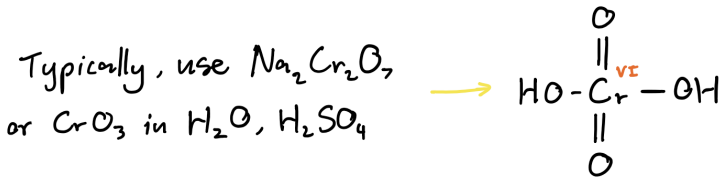

Figure 13: Formation of Chromic Acid Reagent

Chromic acid is best for secondary alcohols, thus resulting in ketones. Primary alcohols, on the other hand, are **overoxidized** in aqueous medium.

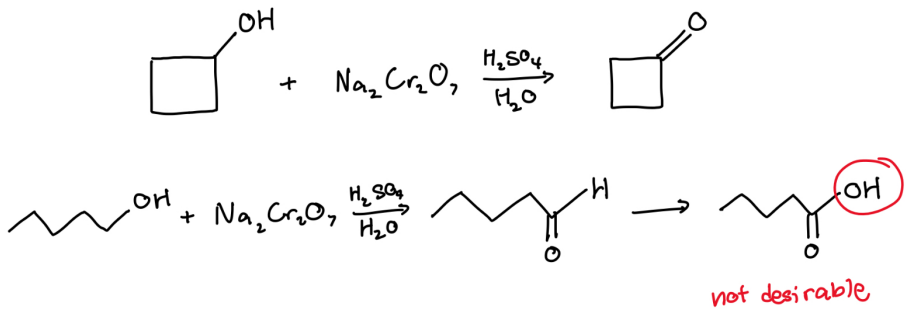

Figure 14: Oxidation of Alcohols via Chromic Acid

We overcome this issue by using pyridinium chlorochromate or PCC which is soluble in organic solvents. This compound avoids the overoxidation that chromic acid would have done.

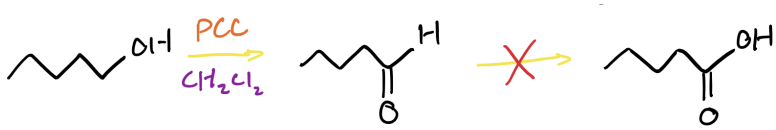

Figure 15: Pyridinium Chlorochromate (PCC) Structure

The third method of alcohol synthesis we will be covering is via **ORGANOMETALLICS**. This method is specifically done through **grignard reagents** (RMgX) and organolithium reagents (RLi). Just as BH_4^- and LiH_4^- are were used to donate hydrides to push the π bond e^-s onto the oxygen, grignard reagents do the same but donates entire R group in the process rather than hydrides.

We can create organolithium reagents via the following.

$$\text{Li} + \text{RX} \xrightarrow[-\text{X}]{\text{hexane}} \text{R} - \text{Li} + \text{LiX}$$

While grignard reagents require ether solvents as we will see soon, organolithium reagents must be done in a nonreactive, nonpolar solvent like hexane. This is due to the EN differences between the two metals.

We can create grignard reagents via the following.

$$\text{Mg} + \text{RX} \xrightarrow{\text{ether}} \text{R} - \text{Mg} - \text{X}$$

This reaction must be done in ether to prevent RMgX from clumping up. Ether allows RMgX to maintain an aqueous state by donating e^- density to the Mg to fulfill its octet (a.k.a. solvation).

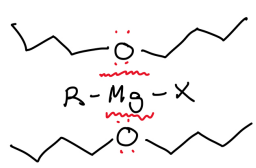

Figure 16: Solvation of Grignard Reagent in Ether

The resulting R – M or R – M – X molecule is one that is polarized because of the EN difference between the R group

and M. However, this time, we see the R group (*carbons*) being the more EN over its cooresponding partner M (because M is more left in the periodic table). This is different from most organic molecules where carbon is usually the one that holds a partial positve charge. Consequently, this makes the carbon basic and nucleophilic.

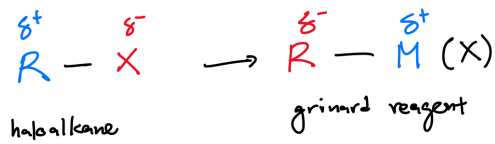

Figure 17: Polarization of Grignard Reagents

This makes them extremely easy to protonate via hydrolysis and thus need to be made by very dry solvents unless the goal was change R – X to an R – H

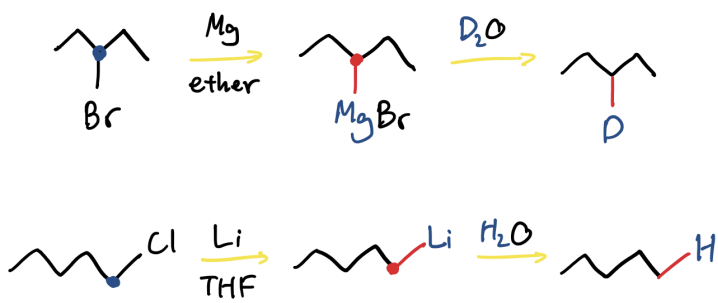

Figure 18: Conversion of Alkyl Halide to Alkane via Grignard

However, when reacted with ketones, aldehydes, or formaldehyde, they form cooresponding alkoxides, which can then be put through an aqueous work-up in order to form alcohol.

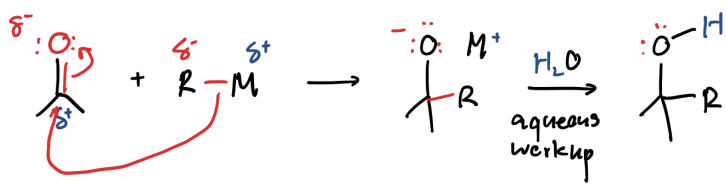

Figure 19: Alcohol Synthesis via Grignard Addition to Carbonyls

This method of alcohol formation is a **very powerful tool** because it allows us to add a R group to any carbonyl containing compound whereas the other methods (such as using BH_4^- and LiH_4^-) could only add H.

3 Alcohol Reactions

As a review of simple acid-base chemistry, in order to deprotonate alcohol, it requires a stronger base than RO^-. In other words, the base must have a stronger affinity to the H^+ than the conjugate base of the alcohol (RO^-). If the alcohol is acidic, water will usually suffice in this matter. However, in cases where water is more acidic than the alcohol itself, H^+ can be donated to the alcohol and H_2O L group will be formed, leaving the resulting product to have a unstable, empty atomic orbital on the C after the L group departs

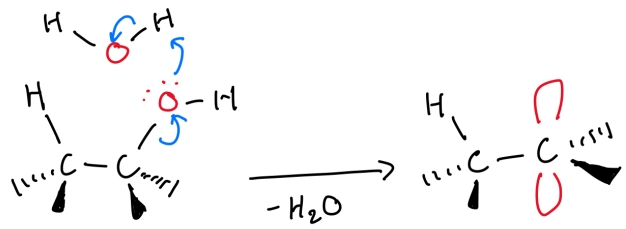

Figure 20: Formation of Oxonium Ion and Carbocation

This phenomenon presents a range of complications when dealing with alcohol reactions. For instance, let's say that we want to replace the $-OH$ with a different functional group like a halide X.

This can occur either through SN2.

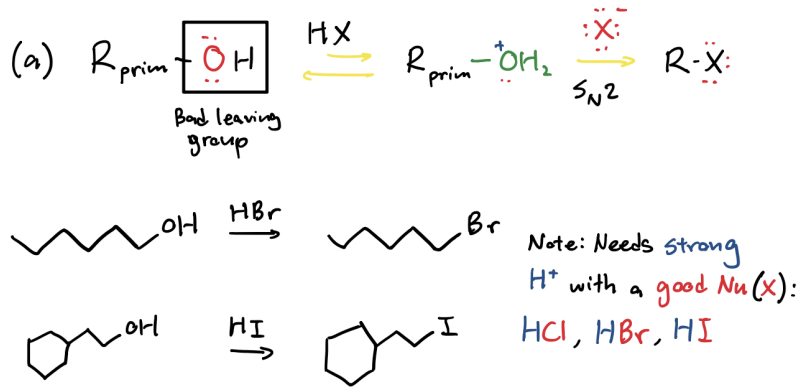

Figure 21: Nucleophilic Substitution (SN2) of Alcohols

Or SN1.

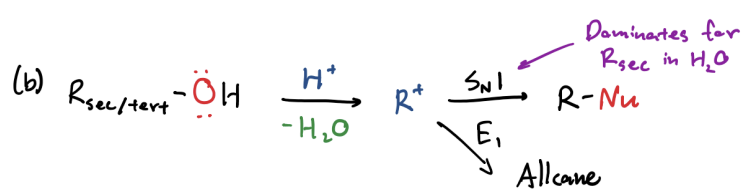

Figure 22: Nucleophilic Substitution (SN1) of Alcohols

However, the issue of mixtures occur via different potential mechanistic occurances.

Firstly, there is the issue of **hydride shifts** during SN1 reactions.

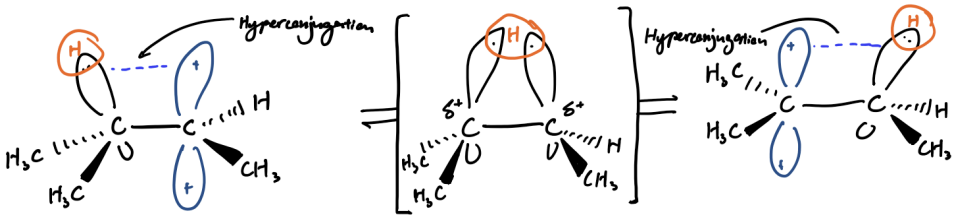

Figure 23: Hydride Shift Mechanism in SN1 Reaction

This is particularly an issue if the hydride shift is a favorable occurance such as the move from a R_{sec}^+ to a R_{tert}^+. Degenerate shifts ($R_{sec}^+ \rightarrow R_{sec}^+$ or $R_{tert}^+ \rightarrow R_{tert}^+$) are less common, but can still occur. This results in the halide attacking at a different location from the one intended.

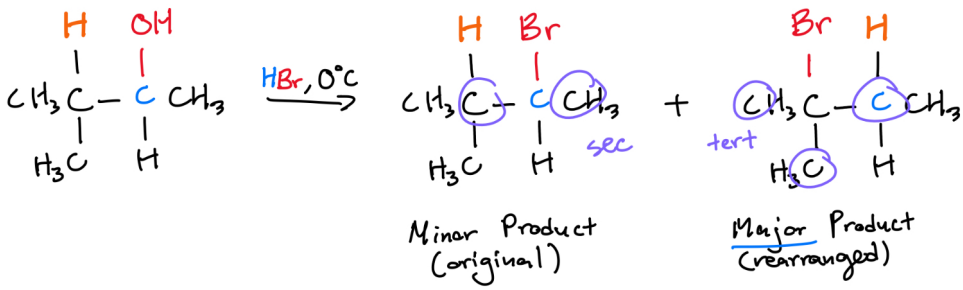

Figure 24: Hydride Shift Resulting in Regioisomeric Products

In specific cases like ring structures, these hydride shifts happen stereospecifically.

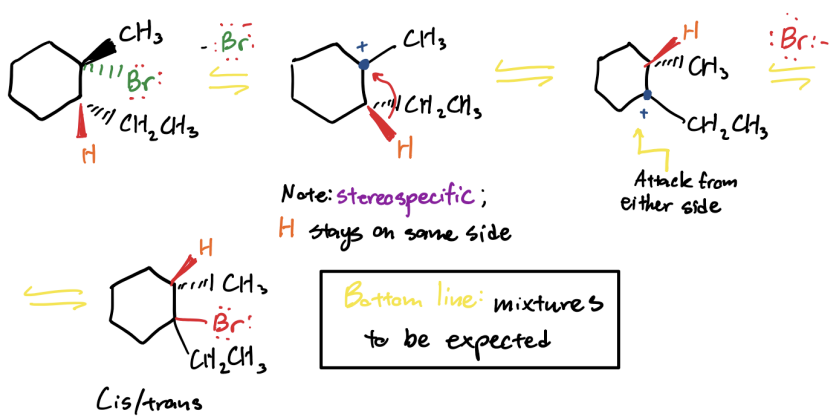

Figure 25: Stereospecific Hydride Shift in Ring Structures

Then, there is the complication of E1 reactions, which become more prevalent at higher T and in absence of a viable nucleophile. *Notice how the H^+ detachment at the last step happens at the tertiary position and not the secondary (where the –OH originally was) due to greater stablity from more hyperconjugation.*

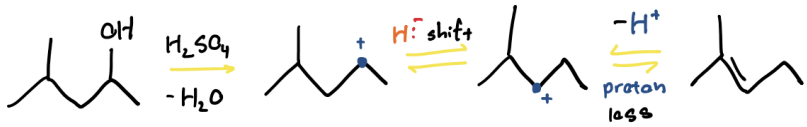

Figure 26: Elimination (E1) Mechanism in Alcohols

Additionally, entire alkyl shifts can occur although not as commonly as hydride shifts because of the greater E_a. These shift can occur even in $R_{prim}-OH$s.

Figure 27: Alkyl Shift Mechanism in Alcohol Reactions

However, it is especially favorable and fast when the shift relieves ring stain as shown in the following figure.

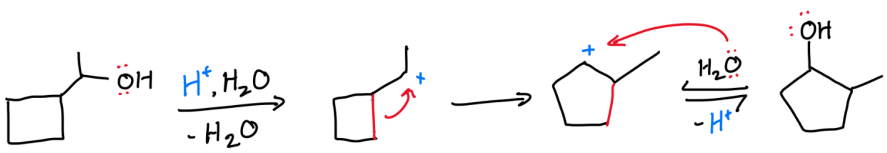

Figure 28: Ring Expansion via Alkyl Shift

So, how do we actually turn an ROH into RNu without all of these potential issues that are bound to happen? The solution is to use inorganic derivatives as reactive or isolable intermediates.

[-OH +Br] Let's first take a look at how to replace an -OH with a -Br in an ROH. We use PBr_3.

$$3\,ROH + PBr_3 \longrightarrow 3\,RBr + H_3PO_3$$

As you can see, one molecule of PBr_3 can be used to change 3 ROH molecules to RBr, resulting in phosphorous acid (H_3PO_3) as a byproduct. The mechanism for this reaction is as follows.

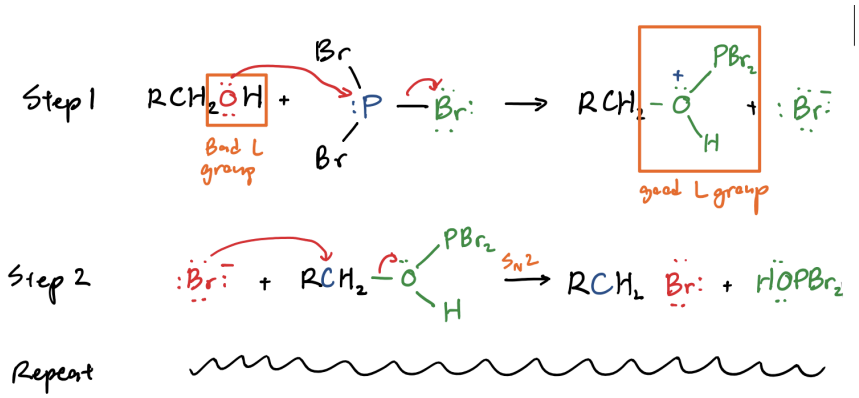

Figure 29: Mechanism of Alcohol Bromination using PBr3

[-OH +Cl] Next, let's take a look at how to replace an -OH with a -Cl in an ROH. We use $SOCl_2$.

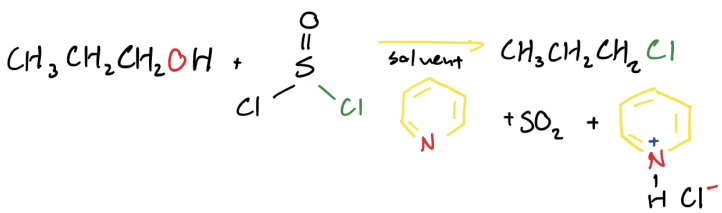

Figure 30: Reaction of Alcohol with Thionyl Chloride (SOCl2)

Again, the mechanism is as follows.

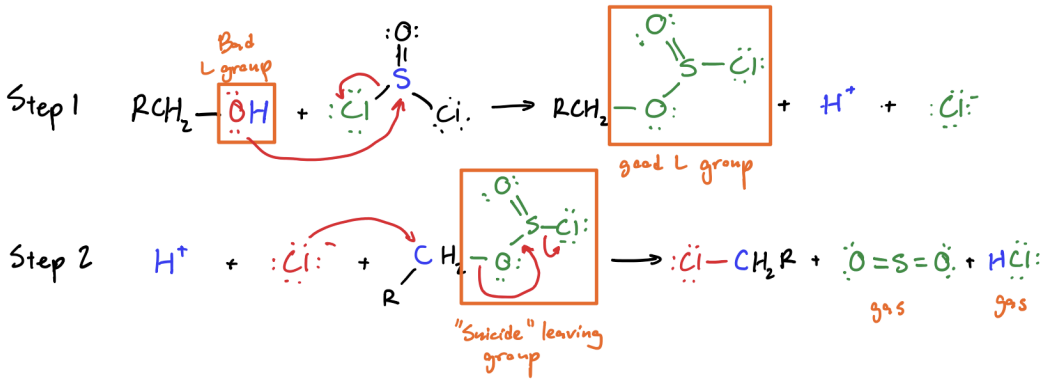

Figure 31: Mechanism of Alcohol Chlorination using SOCl2

[-OH +L] In order to replace an -OH with a good L group, we can use isolable sulfonates such as the following molecules and cooresponding reactions.

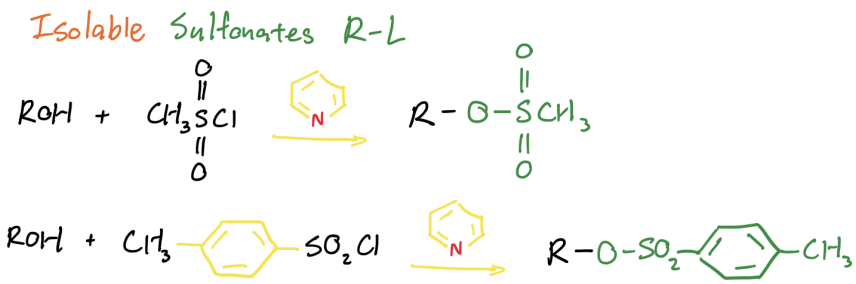

Figure 32: Formation of Isolable Sulfonate Esters

Remember that the displaced Cl^- after the sulfonates attach to the R group is stablized as a floating ion via solvation by the pyradine solvent as seen in the reaction above with $SOCl_2$.